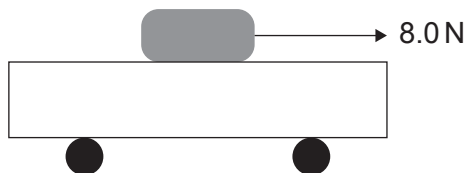


PAPER 1 / QUESTION 56

2 May 2023 / TZ1 (file label) / HL Paper 1 / original question 7 / 2223-6507

56. A block of mass 2.0 kg is placed on a trolley of mass 5.0 kg , moving horizontally. A force of 8.0 N is applied to the block which slides on the surface of the trolley. The frictional force between the trolley and the ground is zero.



The trolley accelerates at a rate of 1.0 ms^{-2} . What is the coefficient of dynamic friction between the block and the trolley?

- A. 0.05
- B. 0.15
- C. 0.25
- D. 0.35